Chapter 2

Terms & Conditions

All non-cited graphs for this course made by me on Desmos

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Prerequired Knowledge for This Class

- This is a 200-level class, but it is a difficult 200-level class

- Calculus is not required/used, but understanding it helps
  - Calculus is for the easy parts of economics anyway

- What is required:
  - Economic terms (compliment, substitute, elasticity, diminishing, etc)
  - Manipulating graphs (quickly reviewed today)
  - Multivariable algebra

Critical Information

- No two people learn the same way
  - No two professors have the same teaching style
  - I try my best to accommodate everyone’s learning style, but I can't read minds
  - Sometimes the best question is “*I don’t understand what we just did, but I also don’t know what question to ask*”

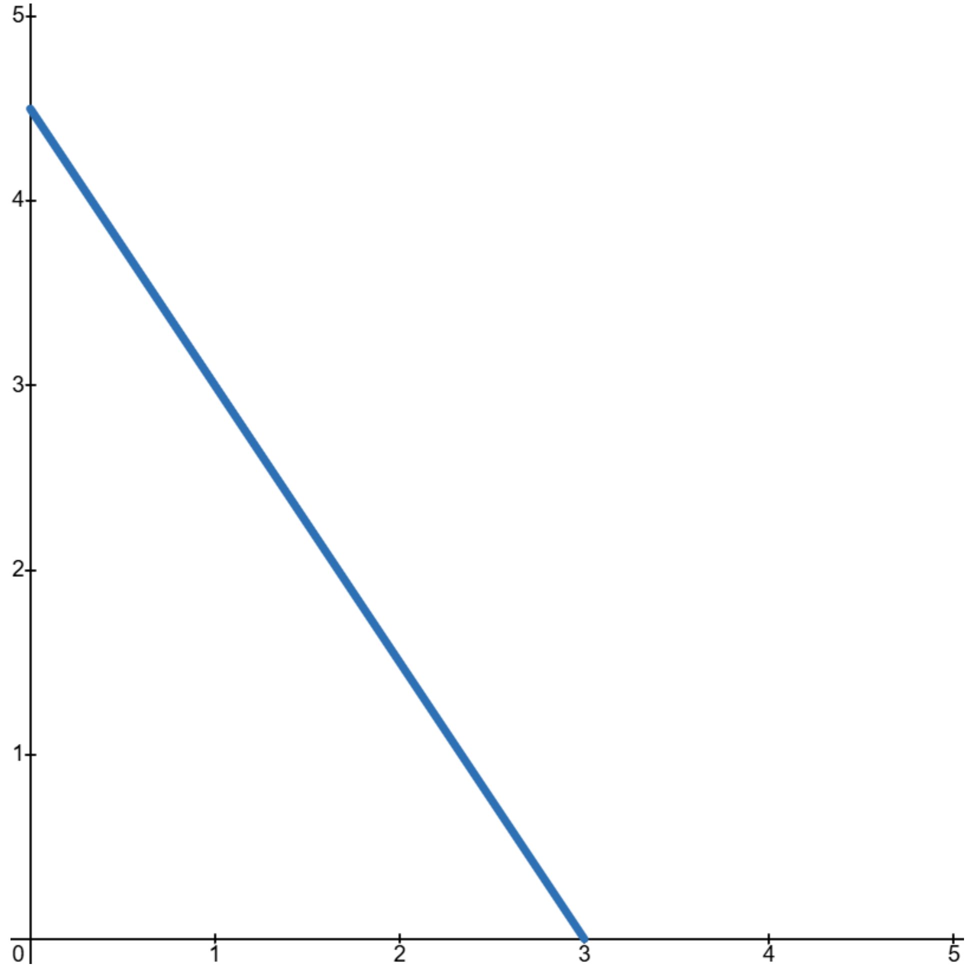
- Discussion isn’t a competition.
  - The only purpose is to learn something new
  - Many times for me, that starts with being wrong

Graphing: lines

› Equation of line: $y = mx + b$
  - y = vertical axis
  - x = horizontal axis
  - m = slope
  - b = vertical intercept

› $y = -1.5x + 4.5$

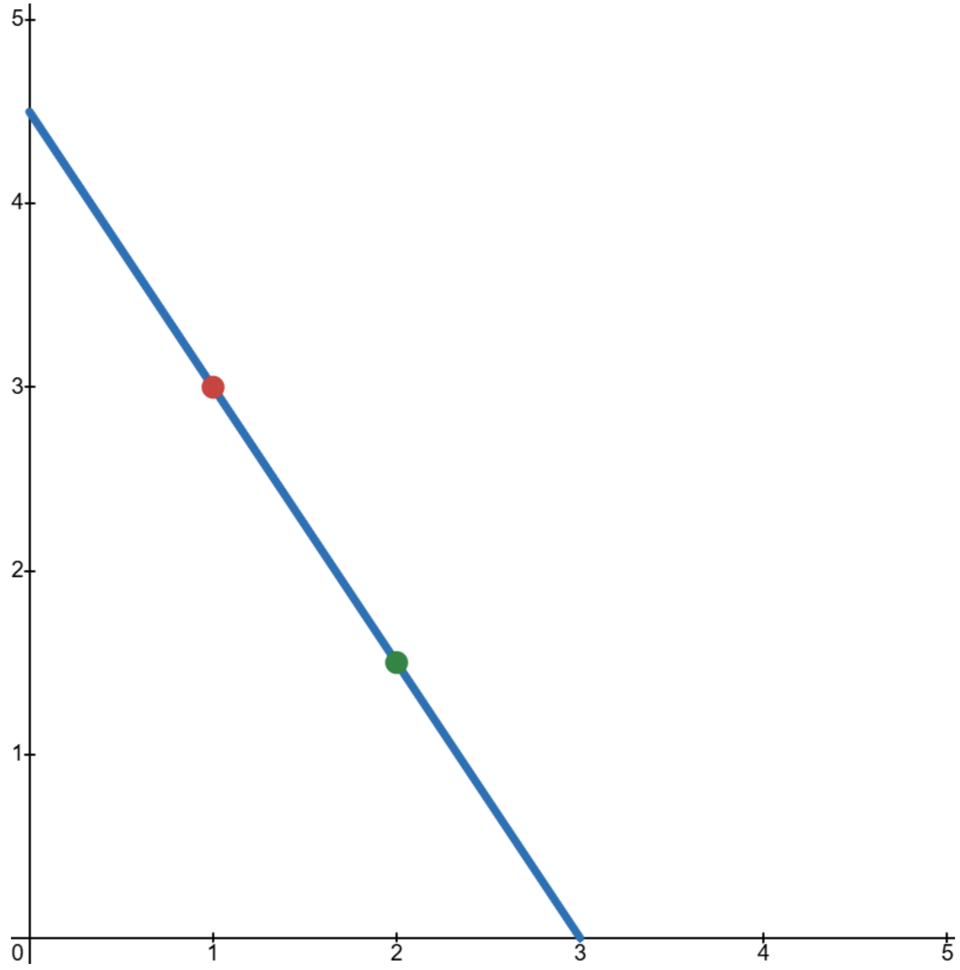
› <u>Slope: change in *y* from changing *x*</u>



Graphing: points

- Points are listed as (*x*,*y*)
  - First point: (*1*,*3*)
  - Second point: (*2*,*1.5*)

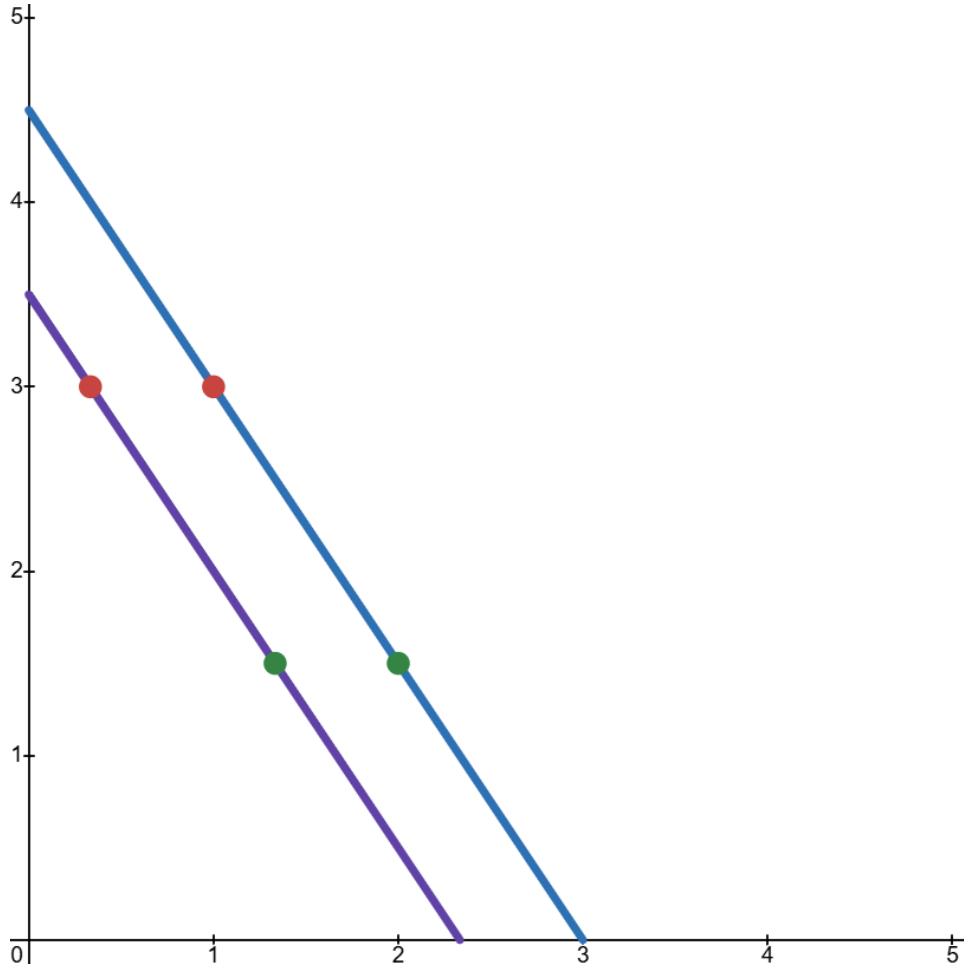
- ***<u>Notice: CHANGING A POINT ON EITHER AXIS DID NOT MOVE THE LINE</u>***
  - Just new (*x*,*y*) on $y = -1.5x + 4.5$



Graphing: shifting lines

- Changing *b* <u>will</u> shift the line
  - Same *y*, but new *x*

- Changes in *b* will have economic interpretation depending on the problem
  - E.g. capital breaks down faster so for a given interest rate, people invest less



Does It Matter How I Draw the Lines?

- Mostly, no. If we explicitly state the slope, then it should be drawn [relatively] straight. If not, either of these is fine:

3 Key Effects for This Class

1. Wealth Effect

2. Substitution Effect

3. Diminishing Marginal Value

Wealth Effect

- **Wealth Effect:** an increase in the present value of your lifetime wealth leads to more consumption of all goods
  - We assume goods are **normal**; demand increases with wealth
  - **Inferior** goods decrease with wealth (ex: fast food)

- Includes future wealth
  - If you know you'll win the lottery in a month, you'll start spending some of that now
  - **Consumption smoothing:** people prefer stable consumption over time, not large jumps

Income Effect

- Similarly, an **Income Effect** will change demand based on current real income
  - If your purchasing power increases, so does demand

- When studying consumption over time, changes in current income can be understood as changes in lifetime wealth
  - We will analyze this through the lens of a wealth effect
    - *My apologies to the supporters of Eugen Slutsky*

Substitution Effect

- **Substitution Effect:** a change in the **relative price** of a good will lead consumers to substitute to the cheaper good
  - Price of one good in terms of another

- Example of relative price:
  - Value of apples = 3
  - Value of oranges = 2
  - Rel. price of apples = 1.5

Diminishing Marginal Value

- **Diminishing Marginal Value:** each additional unit will have a smaller effect on the total
  - Best explain by example. This is saved for Ch. 4
  - Still important to flag now

Economic Agents

- 3 general economic agents:
  1. Households (HHs)
  2. Firms
     - 2.1 Banks mostly work in the background for our class
  3. Governments

- Firms and HHs both have a “problem to solve”
  - Governments studied for policy analysis

Rationality

- The HH's "problem" is to maximize utility, given constraints
  - **Utility** is any benefit from a choice made
  - Called "felicity" in some contexts; roughly akin to happiness
    - Fun thought: if donating to charity brings utility, does altruism exist?

- **Assumption of Rationality:**
  - HHs will try to maximize utility, given constraints
  - Firms will try to maximize value, given constraints

Rationality and Intelligence

› I make no claims about intelligence (for myself included)
  – If dumpster diving for used coffee cups brings you the most utility, then that's what you'll do
  – Utility/value is subjective!!!!

› Gathering information can cost time and money
  – May make some rational decisions appear irrational to others
  – Utility is subjective, including the utility of information

What is a Rational Choice?

› A rational choice will satisfy the <u>first two</u> **choice axioms**

› The **choice axioms** are the foundations of HH preferences
  - Based on relationships of goods (or baskets of goods)
  - Five in total

› Notation used:
  - $\succ$ means "is preferred to"
  - $\succcurlyeq$ means "is at least as good as" or "weakly preferred to"
  - ~ symbolizes indifference between options

Rationality of Preferences

› Axioms for rationality:

1. **Complete preferences:** consumers can make comparisons
   › Ex: apple $\succcurlyeq$ orange
2. **Transitive preferences:** consumers are (basically) consistent
   › If: apple $\succ$ orange, and orange $\succ$ banana, then apple $\succ$ banana
   › No *in*transitive cycling (banana $\succ$ apple)

› Are you allowed to change your mind?

– Of course! Time, thought effort, brain power, etc. are all variables

– Changing a variable changes the outcome

Indifference Curves

- **Indifference curves (IC)** represent utility on a graph

- Important properties:
  1. All points on the curve have the same utility
  2. The higher the IC, the more utility (ordinal)
  3. Infinite number of potential ICs for any combination of goods

- We'll make some assumptions to standardize ICs
  - Different theory classes will look at different preferences
  - Macro generally uses "well behaved" ICs

Continuity Axiom

3. **Continuity:** ICs don't have gaps or jumps
   - If apples $\succ$ oranges $\succ$ bananas, then for a particular λ …
     - (λ)apples + (1-λ)bananas ~ oranges
       - $\lambda \in (0,1)$
   - Rules out Lexicographic preferences
   - (Probably the definition that changes the most when using calculus)

› What does this mean for us?
  - Rational preferences that are continuous can be described with a utility function **(Debreu, 1954)**

[Non] Satiety Axiom(s) (1/2)

› **4a. Local Non-satiation:** there is always a small change you can make that will leave you better-off

- Could be increasing *or decreasing* in quantity
- Ex: having 3 cups of coffee is preferable to having 4

› **4b. Weak Monotonicity:** more is never worse off

- Do I need another book to go unread on my shelf?
- No, but I’m not worse off it one showed up there

[Non] Satiety Axiom(s) (2/2)

- **4c. Strict Monotonicity:** more is always better
  - Rules out perfect compliments (Leontief preferences)

- What does this mean for us?
  - We essentially have 3 goods:
    1. Consumption today
    2. Consumption in the future
    3. Leisure (may occasionally touch on leisure in the future)
  - All else equal, the HH prefers more of all of these

Convexity Axiom(s) (1/2)

- **5a. Weak Convexity:** a combination of 2 goods is at least as appealing as the lesser good
  - If apples $\succcurlyeq$ oranges, then…
    - (λ)apples + (1-λ) oranges $\succcurlyeq$ oranges
      - *All* $\lambda \in [0,1]$

- **5b. Strict Convexity:** same thing, but the combination will always be preferred to the lesser good
    - (λ)apples + (1-λ) oranges $\succ$ oranges
      - *All* $\lambda \in (0,1)$
  - Rules out perfect substitutes

Convexity Axiom(s) (2/2)

- What does this mean for us?
  - Helps us explain the marginal rate of substitution later
  - Final piece that gives IC's their curved appearance
  - *An increase in income means the HH with save some so they can have more consumption today and tomorrow*

- HHs don't only prefer 1 thing
  - Better to have multiple goods, experiences, sources of happiness than only one
  - "*Variety is the spice of life*" **(Cowper, 1802)**

The ‘Macro’ Cheat Sheet of Preferences

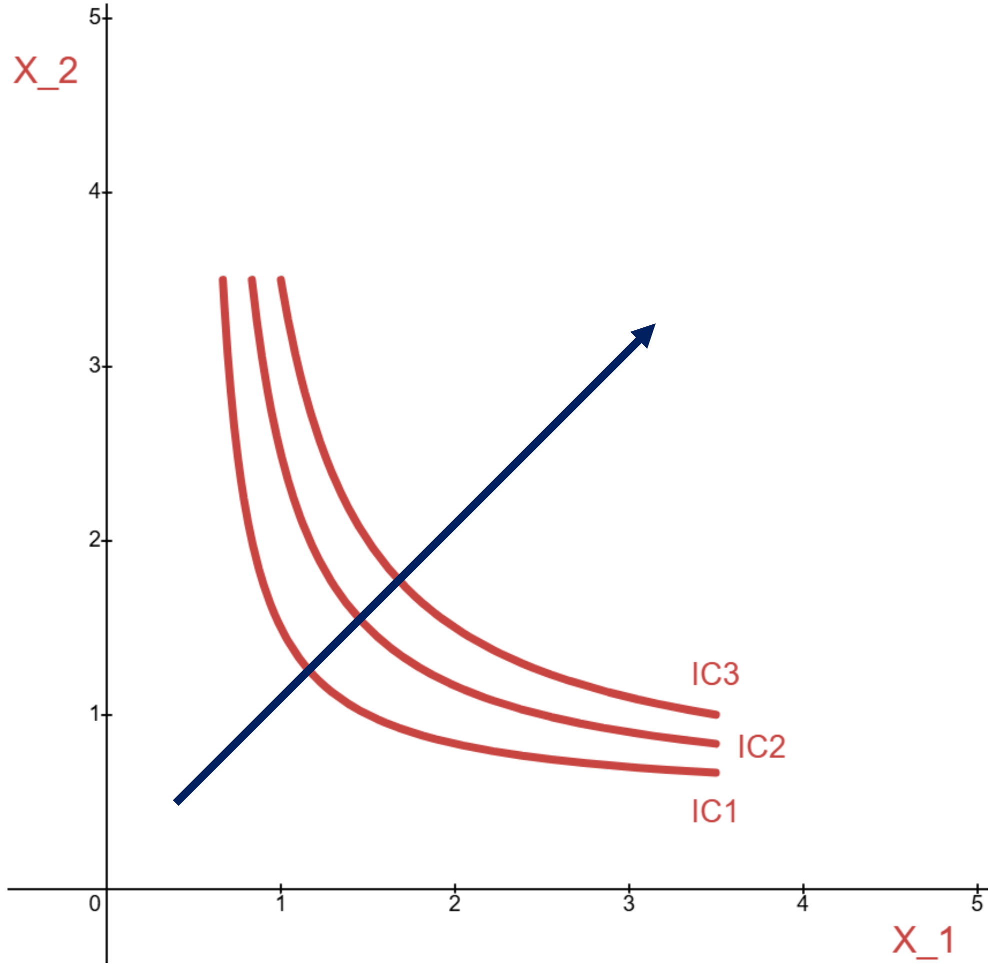
A. Completeness + Transitivity = rational preferences

B. Completeness + Transitivity + Continuity = utility function
   - A and B mostly work in the background for us

C. Strict Monotonicity = HH likes more stuff
   - Solidifies our Wealth Effects

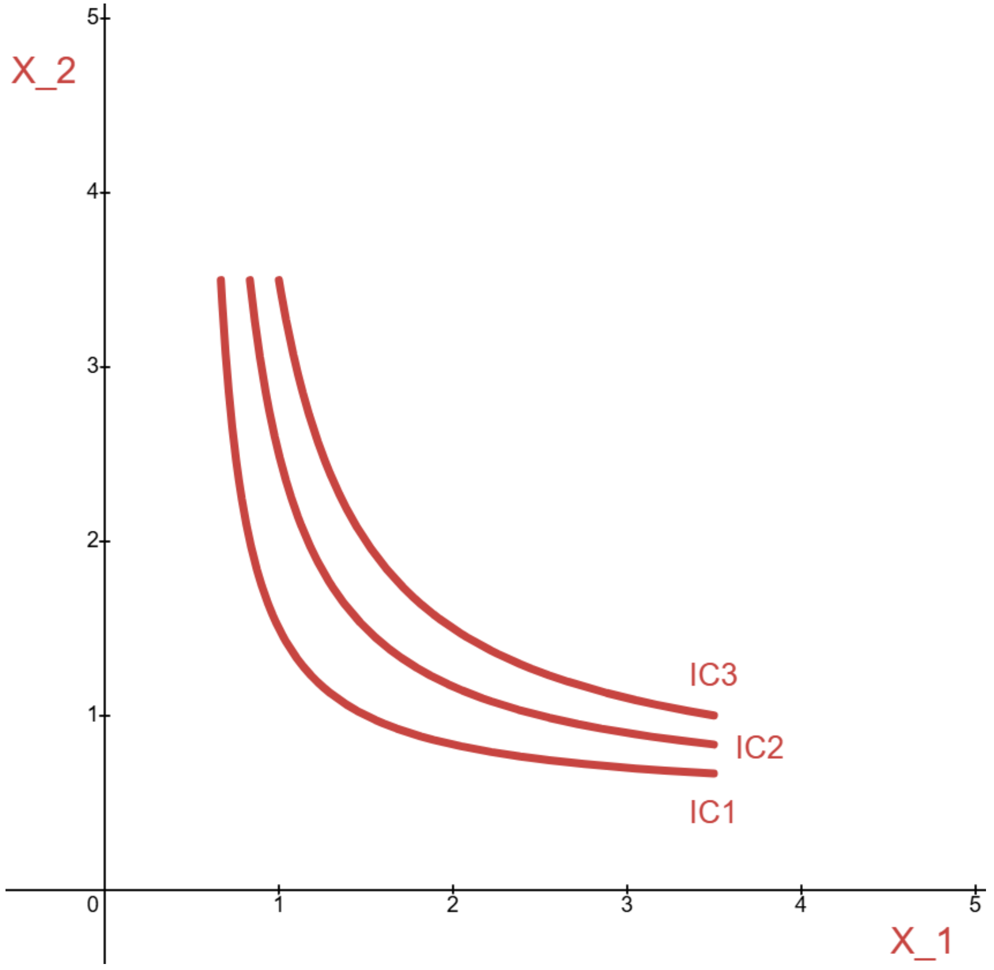
D. Strict Convexity = Combos preferred to single extremes
   - We’ll use this as a “calculus work around”
   - Solidifies consumption smoothing

Our “Well Behaved” Indifference Curves



Satisfying Conditions

- Complete:
- Transitive:
- Continuous:
- Strictly monotonic:
- Strictly convex:



References

1. Cowper, William. *The Task: A Poem in Six Books*. No. 2094. E. Lincoln, 1802.

2. Debreu, Gerard. "Representation of a preference ordering by a numerical function." *Decision processes* 3 (1954): 159-165.

A. All non-cited graphs for this course made by me on Desmos.

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